Jiaxin Li

ljx2986519980@gmail.com | Zurich, Switzerland / China

EDUCATION

ETH Zurich, Zurich, Switzerland Sep 2025 – Present

MSc in Robotics, Systems and Control (RSC)

Courses**:** *Robot Dynamics (6.0), Dynamic Programming and Optimal Control (6.0), Introduction to Mathematical Optimization (6.0), Probabilistic Artificial Intelligence (5.5)*

Shanghai Jiao Tong University, Shanghai, China Aug 2021 – Jun 2025

Bachelor of Engineering in Mechanical Engineering

GPA: 3.91/4.3; Rank: 1/167

RESEARCH EXPERIENCE

Desktop Robot Soccer System with Multi-Agent RL Oct 2024 – Jun 2025

Advisor: Prof. Zhinan Zhang; Shuai Liang | Shanghai AI Laboratory

- Developed a physics-aware simulation environment for a desktop robot soccer platform, enabling reinforcement learning training and evaluation.
- Designed a multi-agent reinforcement learning algorithm with self-play and curriculum learning for cooperative robot soccer.
- Deployed trained policies on real desktop robots, implemented communication and perception in an indoor micro-soccer testbed, and validated real-world system performance.

Health Monitoring with Vibration Sensing and AI Jun 2024 – Sep 2024

Advisor: Assistant Professor Yiwen Dong | UbiHealth Lab, UIUC

- Developed a Gait Phase-Attentive Multi-Task Network to estimate plantar pressure balance from floor vibration data for gait health assessment.
- Designed a joint classification and regression framework to identify abnormal gait patterns and predict three pressure asymmetry targets.
- Achieved 0.85 AUROC and 80% balanced accuracy in held-out subject evaluation, with regression RMSE below 0.06 across all targets.

IMU Bias Impact on Drone Positioning Aug 2023 – Sep 2024

Advisor: Associate Professor Wei Dong | Institute of Robotics, Shanghai Jiao Tong University

- Developed a mathematical model to estimate the upper bound of positioning error in single-anchor drone localization, accounting for IMU sensor measurement bias.
- Showed that the estimated upper bound remained within 2× the actual positioning error along the x, y, and z axes.
- Established that error estimation effectively reflects trends in actual error changes, showing strong consistency.

Biomimetic Butterfly Design | *Associate Project Lead* Sep 2023 – Jun 2024

Advisor: Associate Professor Yanping Lin | Shanghai Jiao Tong University

- Designed a biomimetic butterfly inspired by Papilio (swallowtail) morphology.
- Utilized SolidWorks to develop the mechanical structure and conducted simulations to optimize the proportions of the crank-rocker mechanism and other parameters.
- Developed a final product with wingspan and length up to 30 cm, weighing only 16 grams, capable of stable takeoff and smooth flight.
- Won First Prize in the Shanghai Undergraduate Mechanical Innovation Competition and Third Prize in the National Undergraduate Mechanical Innovation Competition.

COMPETITION EXPERIENCE

China Undergraduate Mathematical Contest in Modeling | *Team Lead* Oct 2023

- Developed a vegetable replenishment pricing decision model based on LSTM and the multi-population genetic algorithm.
- Won the National Second Prize (top 2.3%) and Shanghai First Prize.

RoboMaster | *Member of the Vision Team* Oct 2022 – Aug 2023

- Developed a YOLOv8-based detection model, implementing precise and rapid object detection.
- Applied PnP and inverse kinematics to control a 6-DoF robotic arm for precise positioning and object handoff.
- Won the National Championship of the 2023 RoboMaster season.

Mathematical Contest in Modeling (MCM) | *Team Member* Feb 2023

- Developed ARIMA and LSTM-based models to forecast Wordle user engagement over the following week.
- Won the Meritorious Winner prize for Problem C (equivalent to first prize), placing in the top 7% of global competitors.

INTERNSHIP EXPERIENCE

Shanghai Artificial Intelligence Laboratory | *Drone Team* Jun 2024 – Aug 2024

- Collaborated to develop and iterate a tabletop modular multi-robot platform.
- Performed mechanical modeling, microcontroller programming, and implementation of high-level algorithms.
- Won National Third Prize in the China-US Young Maker Competition.

TEACHING EXPERIENCE

Computational Models of Motion | *Teaching Assistant* Spring 2026

- Created programming assignments for approximately 120 students on trajectory optimization for bipedal robots.
- Covered reference trajectory generation, kinematics, optimization, and simulation in the assignment design.